Problem Chosen	2024 MCM/ICM Summary Sheet	Team Control Number
E		2422375

Catastrophes Coming , Insurances Emerging

Summary

"When we have a fact-based worldview, we can see that the world is not as bad as it seems-and we can see what we have to do to keep making it better."—-Hans Rosling.

High incidence of extreme weather threatens not only our lives and property but also increase the insurance protection gap. It is a delimma but also an opportunity for the insurance industry to step up in times of crisis. Therefore, we developed an **Insurance Program Forecasting Model (IPFM)**, **Customer Site Selection Model(CSSM)** and a **Landmark Preservation Model (LPM)**to control the cost and value in extreme weather.

For task 1: To ensure the sustainability of insurance companies in the face of catastrophes, we developed the **IPFM model**. First, we used ArcGIS to screen out the types, degrees, and frequency of extreme weather, and we compiled statistics on insurance companies. Then, we used **Integer programming** in comprehensive risk loss ratio, profit, and company operating costs. Then, we used a **Logistic Regression** to determine when the insurance companies will underwrite during a catastrophe. The model is tested to be **0.85** accurate. And the reinsurance program is the choice of the insurance company to come forward during a catastrophe. Finally, we selected two locations, England and Southern California, USA to demonstrate our model and propose corresponding insurance programs.

For task 2: The increasing catastrophe risk also makes the property developers face new challenges. Thus, we developed a model based on **Genetic Algorithm (GA)** optimization and **Prim Algorithm**. First, we selected five major factors that affect real estate site selection: population, transportation, competition, environment, and **IPFM**. Then, we use Prim to ensure the lowest cost after selecting all factor points. Finally, the GA was used to optimize the solution process.

For task 3: We built **LPM** to assist community managers in developing a landmark preservation plan in High-risk areas. First, we established a landmark value evaluation system based on the **COWA-Grey Clustering Method**. We selected Four primary indicators: Culture, History, Economy, Community, and their secondary indicators. Then, we determine their weights by the COWA operator assignment method and use the Grey Clustering Method to obtain the landmark value index. Besides, a Value-Cost analysis model based on the **GE matrix** categorizes preservation options into three categories to assist leaders in making decisions. Eventually, we applied the model to Cape Hatteras Lighthouse and the Statue of Liberty and proposed protection strategies. Their landmark value index is 7.695 and 7.967.

Meanwhile, we conducted a sensitivity analysis of the LPM by changing the parameters of cost. The results proved the practical feasibility of our model.

For task 4: We applied insurance IPFM and LPM to St. Mark's Basilica in Venice, Italy, and proposed a preservation plan, timeline, and cost proposal in a letter to local community leaders.

Keywords: IPFM;GA-Prim;COWA-Grey Clustering;GE matrix

Contents

1 Introduction

1.1 Problem Background

Radical weather is fast becoming an existential crisis for property owners and their insurers. In 2021, in the United States, natural disasters, represented by tornadoes, tropical storms, and deep freeze, caused economic losses of roughly US 145 billion dollars. In Europe, torrential rainfall triggered severe floodings that caused overall losses of about US 54 billion dollars. In the face of huge economic losses, purchasing insurance is an important way to reduce the pressure of rebuilding. In 2021, natural disasters caused total losses of 280 billion dollars, of which about 120 billion was insured.

Insurance companies can fulfill their responsibilities by assuming some of the risks and losses. Applying a risk-adequate premium prices natural disasters and limits losses. But as disasters become more severe, insurers are also facing a crisis of a widening insurance protection gap: the emerging crisis in profitability for the insurers and in affordability for the property owners.

To reduce future losses to the community, we need to develop sound insurance strategies. Maximize the interests of the protection company and our clients. Realize the social and commercial value of insurance. At the same time, we need to assist leaders in areas at high risk of disaster to develop appropriate landmark protection strategies. Sustainability is our shared model.

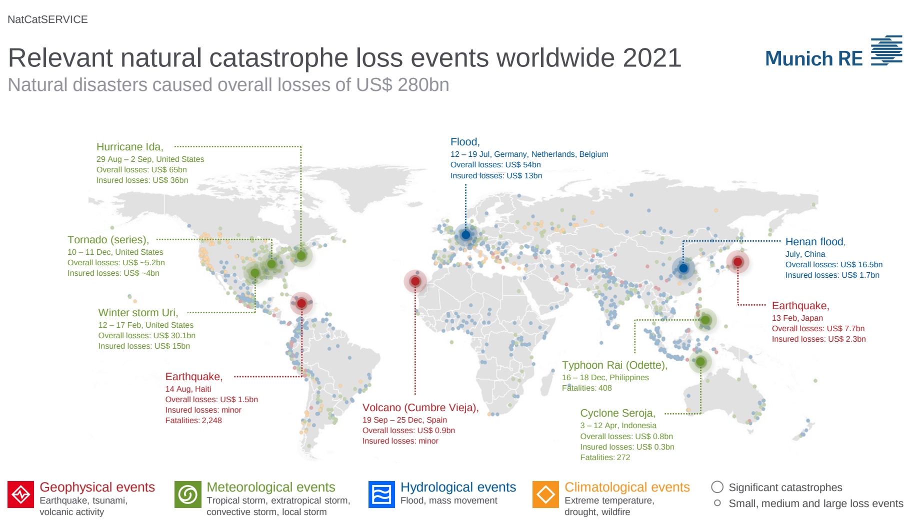


Figure 1: Revelant natrural catastrophe loss events worldside 2021

1.2 Restatement of the Problem

Considering the background information and restricted conditions, we need to solve the following problem:

- Problem 1: Building an insurance predicting model to help insurers measure the risk of extreme weather hazards in the region. Determine a reasonable insurance underwriting program. Select two regions to practice.

https://www.munichre.com/

- Problem 2: Taking into account the above insurance options and factors, we need to assist communities and property developers in determining how and where to build and grow. It needs to adapt to a growing population, climate change, and social needs.
- Problem 3: For some areas where the risk is extremely high and insurance is denied, Establish a landmark building protection model. It includes determining evaluation indicators and establishing an evaluation system for the value of landmark buildings. Analyse and formulate reasonable protection strategies.
- Problem 4: Choose a historic landmark and use our model to analyze it. Write a letter to the community leader, recommending our plan.

1.3 Our Work

The overall flow of the work we have done is shown in Figure 2.

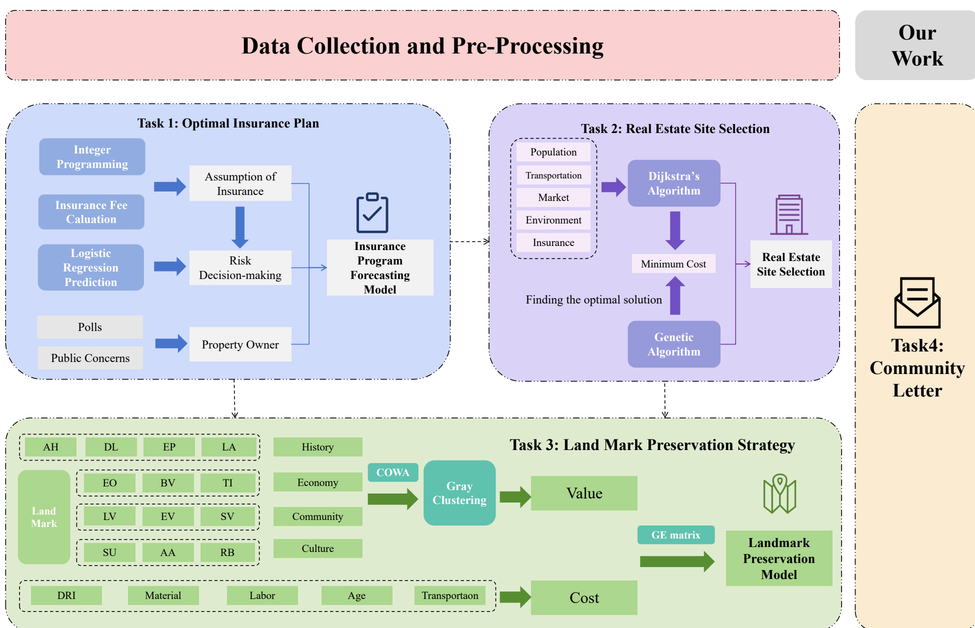


Figure 2: Flow chart of our work

2 Assumptions and Explanations

▼ **Assumption1**: All the data used in the article is valid.

▲ **Explanations**: Our data comes directly from the databases of reliable official websites and the latest study

▼ **Assumption2**: The price of premium payouts determined is independent of insurance selling means.

▲ **Explanations**: When determining the price of insurance payout, we only consider objective factors such as the operating costs of insurance companies and the risk of natural disasters caused by extreme weather. The selling method is out of our concern.

▼ **Assumption3**: The optimal insurance solution maximizes the benefits for both the insurance company and the insured area.

▲ **Explanations**: We need to maintain the number of customers of the insurance company, so the cost of insurance should not be too high. To reduce the risk, we need to strike a reasonable balance between compensation economic loss of the disaster, reducing the Insurance Protection Gap, and maximizing the interests of both parties.

▼ **Assumption4**: The changes in the financial environment are not taken into account.

▲ **Explanations**: Costs associated with the analyses in this paper are calculated in present-value dollars. Future projections and inflation adjustments are out of our consideration, as it is difficult to predict these changes over the life of the insurance policy. Additional assumptions are made to simplify analysis for individual sections. These assumptions will be discussed at the appropriate locations.

3 Notations

Table 1: Notations Used in this Paper

Symbol	Description
K_i	The disasters loss ratio
DRI	The risk level of Disasters
Y_i	The direct economic losses from disasters
R_{DRI}	The pure premium rate of the insurance
P_{DRI}	The probability of different loss ratios of disasters
L_{DRI}	The insurance premium
T_i	The company's value
Q	The insurance amount
C	The insurance amount for compensating different risk levels
η_i	Parameters of the cost formula
U^T	Evaluation thresholds

• Other symbols will be given in the following text.

4 Insurance Program Forecasting Model (IPFM)

4.1 Calculation of Loss Ratio and Premium for Extreme Natural Disasters

We aim to determine the conditions under which insurance companies assume the policy, ensuring their interests while effectively dealing with the impact of extreme climate change.

Initially, we conduct a correlation analysis on the identified catastrophe insurance (CAT, ILS) and the corresponding catastrophe risk data. The analysis reveals catastrophe insurance mainly correlation with storms in the United States and floods in Europe, while the correlation between disaster-prone regions in Asia and insurance is not significant.

Consequently, we select the Total Damage Dollar (Y_i) from the catastrophe data to represent the extent of catastrophic damage.

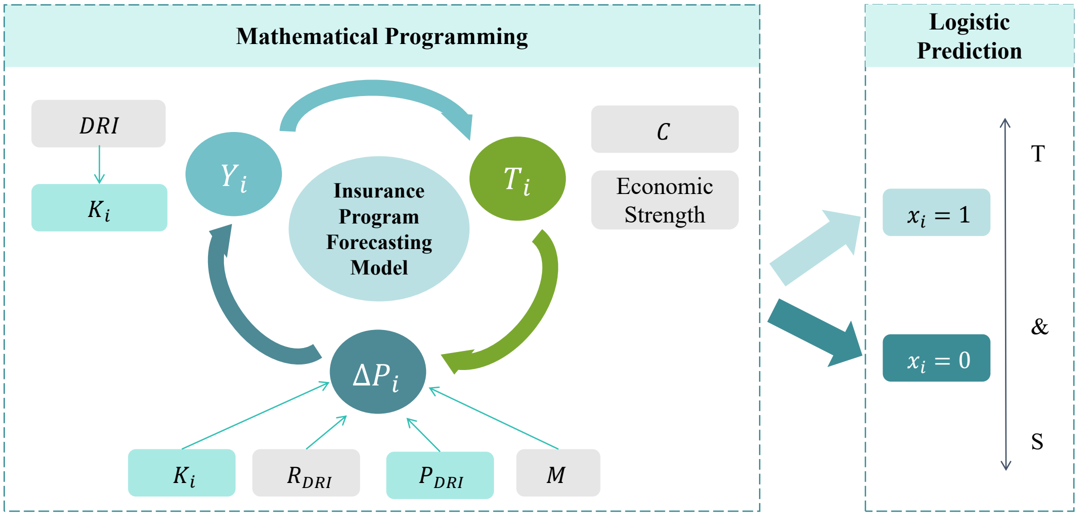


Figure 3: IPFM Flowchart

$$K = \frac{Y_i}{Y_{GDP}} \times 100\% \tag{1}$$

Among them, K means the catastrophe loss ratio. Y_i means the direct economic losses from catastrophes. Y_{GDP} means the gross domestic product of the affected country or area.

Additionally, through an analysis of the factors of catastrophe losses and the composite risk index, we have found a linear correlation between the composite risk index of catastrophes and the direct economic losses caused by catastrophes. Therefore, the higher the Disaster Risk Index of catastrophes (DRI), the higher the resulting economic losses.

$$K = 0.0002DRI - 0.0028 \tag{2}$$

According to the theory of expected loss, the pure premium rate R of catastrophe insurance can be obtained by dividing the expected loss by the actual loss amount. The expected loss is the sum of the product of the catastrophe loss ratio and the probability of catastrophe loss risk. Therefore, the formula for calculating the pure premium rate of this insurance is:

$$R = \frac{E\left(Loss\right)}{\lambda Y_i} \tag{3}$$

Among them, Y represents the actual loss value of disasters, and $E(Loss)$ represents the expected loss value.

The degree of protection of flood insurance for policyholders is not specified. The pure premium rate is the main component of the insurance rate. It refers to the ratio of the pure premium to the insurance amount, and this portion of the pure premium will be used for compensating and paying the insured after an insurance accident occurs[2].

The rate R_{DRI} of catastrophe insurance is the sum of the product of the catastrophe loss ratio K and the probability P of losses caused by catastrophes.

$$R_{DRI} = \frac{E\left(Loss\right)}{\lambda Y_i} = \sum_{i=1}^{n} P_{DRI} \times K_i \tag{4}$$

We define K_i as the different loss ratios of different catastrophes, and P_{DRI} as the probability of different loss ratios of catastrophes. The risk level of catastrophes is represented by the DRI, and since the risk level corresponds to the loss ratio, the probability of different loss ratios of catastrophes is the probability of corresponding level catastrophes occurring. The premium of insurance is the product of the insurance rate and the insurance amount, calculated by the formula:

$$L_{DRI} = R_{DRI} \times Q \tag{5}$$

Where L_{DRI} means the insurance premium, and R_{DRI} means the pure premium rate of the insurance. Q means the insurance amount.

We classify the comprehensive risk of extreme natural disasters into five levels, with different loss rates for different levels of extreme disaster risks.

4.2 Modeling the Insurance Program Forecasting Model (IPFM)

4.2.1 Using Integer programming for the Underwriting Options

Using integer programming, we can choose the best time to underwrite. The above equation indicates that insurance companies decide whether to assume insurance policies based on whether the customers are insured and whether the area is affected by disasters. We assume that insurance companies have the following conditions for assuming insurance policies:

The minimum net income of the insurance company is greater than 0.

The insurance company must ensure its operational costs when assisting customers.

Without considering special human factors such as illegal activities, the insurance company and the affected area seek to maximize their interests.

The process of dynamic mathematical programming is as follows:

$$x_i = \begin{cases} 0, (Notcovered) \\ 1, (Underwirting) \end{cases} i = 1, 2....n \tag{6}$$

$$P_i = M_i \times R_{DRI} - \sum_{i=1}^{n} P_{DRI} \times C_i \tag{7}$$

where M_i represents the premium for each customer, R_{DRI} means the pure premium rate of the insurance, P_{DRI} means the probability of different loss ratios of catastrophes, and C_i represents the insurance amount for compensating different risk levels of disasters by the insurance company.

$$s.t. \begin{cases} \sum_{i=1}^{n} x_i Y_i + P_i + P_\alpha < T_i \\ \min P_i + P_\alpha > 0 \\ x_i \in \{0, 1\} \end{cases} \tag{8}$$

where x_i means whether to assume insurance policies, P_i means the profit from catastrophic insurance, Pa represents the profit from other insurance products of the insurance company, Y_i represents the direct economic loss caused by extreme natural disasters, and Ti means the company's value.

And we need to prevent the insurance company from going bankrupt in case of losses and to ensure the normal operation of the insurance company with a minimum total profit greater than 0.

$$F_{(x)=} \max \sum_{i=1}^{n} P_i x_i \tag{9}$$

Using Matlab, we can determine under what conditions the insurance company chooses to assume insurance policies to maximize its benefits.

4.2.2 Using logistic regression to predict whether to underwrite

The strategy of whether to assume different insurance policies can be obtained through the above equation. By using logistic regression, we can predict the future based on existing data. The specific steps are as follows:

First, the model constructs a linear combination of independent variables as follows:

$$z = \mu + \alpha_i Weather + \beta_i Country + \gamma_i Probit + \eta_i Damdge \tag{10}$$

Then, the linear combination result is transformed into probabilities using the logistic function:

$$P\left(X=1\right) = \frac{1}{1+e^{-(\mu+\alpha_i \text{Weather}+\beta_i \text{Country}+\gamma_i \text{Probit}+\eta_i \text{Damdge})}} \tag{11}$$

Then, the logistic model is transformed as follows:

$$\ln \tfrac{\pi}{1-\pi} = \quad \mu + \alpha_{_i} \text{Weather} + \beta_i \text{Country} + \gamma_i \text{Probit} + \eta_i \text{Damdge} \quad 0 < \pi < 1 \tag{12}$$

Next, we need to perform maximum likelihood estimation on the parameters of this logistic regression.

For this model, we have:

$$P\left(X=1\right) = 1 + e - \left(\mu + \alpha i Weather + \beta i Country + \gamma i Probit + \eta i Damdge\right) \tag{13}$$

The probability that the insurance company will underwrite it.

$$P\left(X=0\right) = 1 - P\left(Y=1\right) \tag{14}$$

For a given dataset, the likelihood function is the product of the probabilities of all observed data under the current model parameters.

$$L\left(\mu, \alpha_{\ _i}, \beta_i, \gamma_i, \eta_i\right) = \prod_{i=1}^{n} \left[P\left(Y=1\right)\right] Xi \left[1 - P\left(Y=1\right)\right] 1 - Xi \tag{15}$$

In this case, xi represents whether the ith policy chooses to be insured (0 or 1), X is the independent variable, and β is the model parameter.

Maximizing the likelihood function: By adjusting the model parameters$\mu, \alpha_i, \beta_i, \gamma_i, \eta_i$ coefficientsour goal is to find a set of parameters that maximize the likelihood function. This is typically achieved through iterative algorithms such as gradient descent.

Specific parameters $\mu, \alpha_i, \beta_i, \gamma_i, \eta_i$can be obtained and model verification can be conducted using SPSS.

4.3 Practices for property owner

It is advisable to consider property insurance, such as home insurance, within one's economic means.

Faced with the economic losses caused by catastrophic events, ordinary property owners can also choose suitable insurance plans to protect their assets. For example, we can consider home and automobile insurance from major insurance companies within our affordability. Additionally, purchasing personal accident insurance can ensure personal safety.

For homeowners in areas prone to floods, hurricanes, and other natural disasters, we can choose to live in homes constructed with more sturdy materials and advanced building techniques. It is also important to assess the disaster prevention and mitigation measures in the vicinity of the residence, with particular attention to the community's emergency response capabilities.

When the local risk of natural disasters is extremely high, we can consider choosing safer areas to live. By continuously improving our risk awareness, we can better protect our lives and property.

4.4 Selection of demonstration areas in South Carolina and the England

4.4.1 Data preprocessing

Economic losses are linearly correlated with the comprehensive loss index. Therefore, we reclassify the economic losses caused by extreme weather disasters into five categories:

Table 2: CRI level

Composite risk index level	1	2	3	4	5
K Loss rate(USA strom)(E-07)	0-0.3	0.3-1.2	1.2-2.0	2.0-15.9	Above 16
K Loss rate(UK flood)(E-07)	0-0.1	0.1-0.3	0.3-1.1	1.1-2.8	Above 2.8
P_{DRI}USA	0.03	0.04	0.11	0.35	0.47
P_{DRI}UK	0.12	0.12	0.19	0.24	0.33

4.4.2 Case Study of the South Carolina and the England

According to these data, we applied our logistic regression prediction model to both the South Carolina and the England. The likelihood ratio chi-square test results showed significant values (P-values) of 0.000*** and 0.001*** for the South Carolina and the England, respectively. Both values are much smaller than 0.05, indicating a significant level of significance and rejecting the null hypothesis, thus confirming the effectiveness of the model.

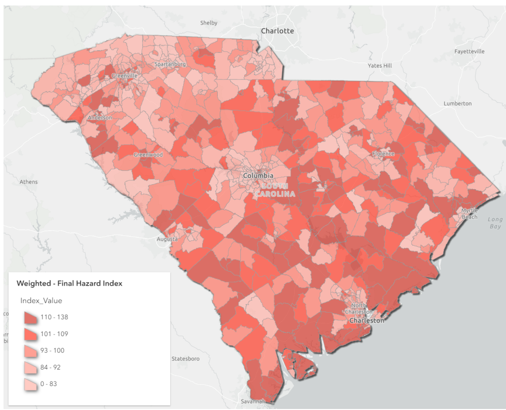
To further quantify the performance of logistic regression classification, we used classification

www.artemis.bm,www.noaa.gov

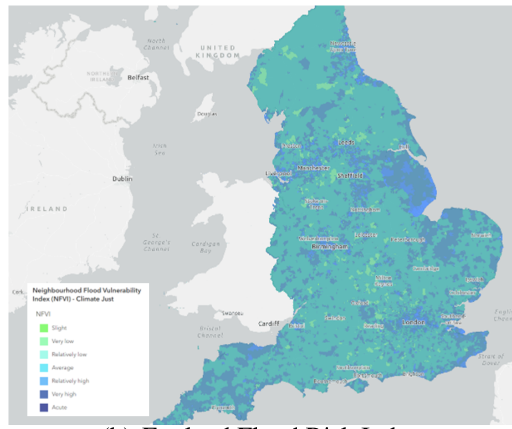
evaluation metrics.

Table 3: Further quantify the performance of logistic regression classification

Area	Accuracy	Recall	Precision	F1	AUC
South Carolina	0.853	0.853	0.859	0.855	0.964
England	0.867	0.867	0.871	0.868	0.977



(a) SC Hurricane Risk Index



(b) England Flood Risk Index

Figure 4: Risk Index

The combined values of various indicators indicate that the predictive performance of the established multinomial logistic regression model is above average and has a certain significance.

Our model can effectively assist insurance companies in different states and regions in deciding whether to provide coverage based on geographical and climatic factors.

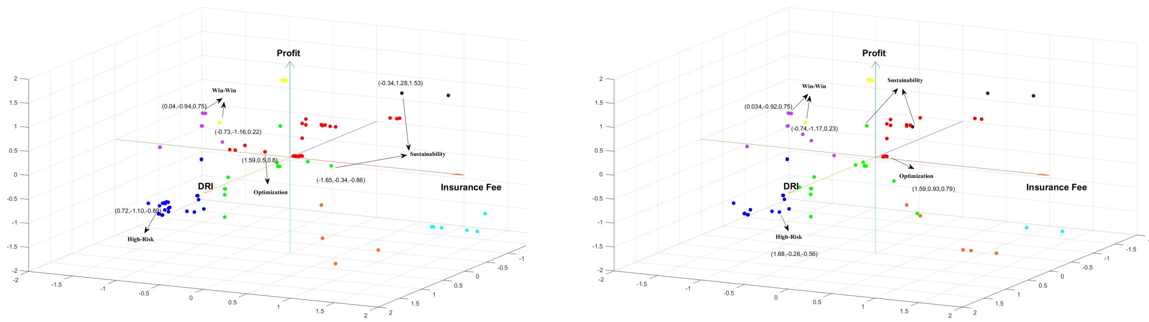
The final linear combinations are as follows:

$$z_{SC} = 50.7 - 19.2Weather - 28.9Country - 27.4Probit - 23.8Damage$$
$$z_{England} = 42.6 - 23.6Weather - 24.1Country - 30.6Probit - 27.4Damage$$

The insurance policy model was applied to the South Carolina and England regions. By considering the insurance risk, insurance company loss-profit, and calculated insurance premiums, a three-dimensional evaluation graph was created to visually display the insurance plan choices under different scenarios.

1. Win-win Strategy: Modest Profit - Effective Underwriting

For regions with lower disaster risk and economic loss, insurance companies can reduce premiums when formulating insurance strategies. This can alleviate the pressure of regional risk protection while still ensuring profits. For instance, the purple and yellow clusters have achieved the win-win strategy in our model.The insurance strategy evaluation graphs for the regions in the two countries are shown below:



(a) South Carolina (b) England

Figure 5: 3D Insurance Scenario Decision Map

2.Low-Risk Sustenance Strategy: Aim for No Loss - Pre-Disaster Insurance

For areas with lower risks, insurance companies can develop low-premium protection plans to attract clients. Regions suitable for such plans can maintain local real estate, public assets, company properties, etc., through purchasing lower-priced insurance. Meanwhile, insurance companies can also develop new clients, gaining momentum for sustainable growth. As illustrated by the black and green point clusters.

3. High-Risk Underwriting Strategy: In Times of Peril, Responsibility Prevails - Catastrophic Reinsurance

The red point cluster, representing areas with higher disaster risk, medium premiums, and low company profits, symbolizes this optimized insurance plan. It also significantly protects regions prone to natural disasters. This ensures that regions do not have to bear excessively high premiums for disaster risks.

4.Non-Underwritten Strategy: Disasters with Over 100 percents Coverage Rate - Powerless Plan

In fact, the economic losses caused by major extreme disasters are often substantial. For high-risk areas, the frequency of disasters is also higher. This plan enables insurance companies to operate normally, laying the foundation for sustainable development. At that moment, insurance companies may also choose to refuse underwriting to preserve their costs. As shown by the blue point cluster.

Due to the high premiums set and the insurance company's low estimated earnings, the insurance company may face significant compensation pressure, and the region could experience substantial economic stress. Therefore, the insurance company might consider refusing to underwrite. These regions can seek assistance from the government, charitable organizations, or crowd-funding to protect local significant assets.

For the South Carolina, the primary risks of extreme weather disasters stem from hurricanes, cold waves, etc., affecting many areas. Simultaneously, the U.S. has a mature catastrophe insurance industry and a high level of economic development, allowing many areas to choose the High-Risk Underwriting Strategy. For some areas severely affected by disasters and holding significant local assets, considering submitting higher insurance premiums is advisable. Facing catastrophic events, insurance companies can also appropriately concede profits.

For the England region, the primary extreme weather disasters faced are mainly floods, etc. Its catastrophe insurance coverage is limited, with most still concentrated in the Low-Risk Sustenance Strategy. When facing catastrophes, European regions need to consider collaborating with governments to mitigate the risks brought by disasters.

5 Customer Site Selection Model(CSSM)

5.1 Selection of indicators

First, we determined the optimal construction location for property developers by selecting the weights of five factors. Thenwe identified secondary indicators through search engines and literature[4].

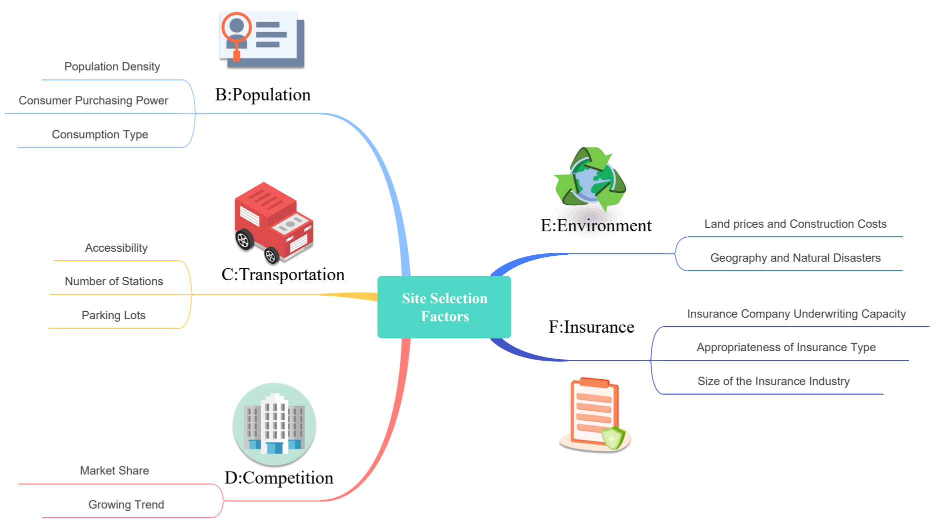


Figure 6: GA-Prim Flowchart

In addition to the considerations typically made by real estate developers in aspects B, C, and D, we also consider the risk index of natural disasters in the environment and the cost impact of disaster-resistant buildings in the area. Furthermore, we consider insurance factors, as reflected in the changes in insurance plans by insurance companies in times of frequent extreme natural disasters, real estate developers also need to consider the insurance plans provided by insurance companies to increase their profits and minimize their losses.

5.2 The modeling process of GA-Prim algorithm

Combining the IPFM in the first question with the insurance plan, the second question utilizes the genetic algorithm-optimized Prim algorithm to find the minimum spanning tree from a graph theory perspective. The minimum spanning tree refers to minimizing the cost loss for real estate developers. Using a genetic algorithm to solve an optimization problem means there are many solutions x to this optimization problem. The optimization process aims to find such x that minimizes the corresponding fitness function f(x).

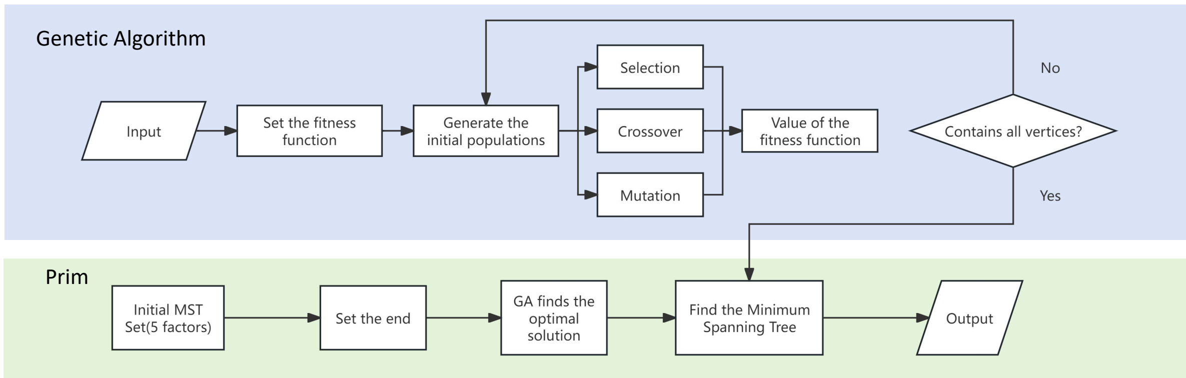


Figure 7: GA-Prim Flowchart

5.2.1 Chromosome Encoding

We naturally sorted the vertices in the graph by vertex number. Each candidate vertex is then encoded as a gene in the chromosome. A gene value of 1 indicates that the path selects the corresponding vertex, while a gene value of 0 indicates no selection. The arrangement of the genes in the chromosome represents the order in which the vertices appear in the path. The length of the chromosome should be equal to the number of vertices in the graph.

5.2.2 Fitness Function

For a graph with n vertices, knowing the edge length $d\left(v_i, v_j\right)$ of each vertex $\left(v_i, v_j\right)$, define the path length of a pathway v_{i1} to v_{jn} $v_{i1}, v_{i2}, \cdots, v_{in}$ as an adaptation Function.

$$f(i)=\sum_{r=1}^{n-1} d\left(v_{ir}, v_{ir+1}\right) \tag{16}$$

For this optimization problem, it is to find the solution x_n that minimizes the value of $f\left(x_n\right)$.

5.2.3 Selection Operation

The selection of parents for crossover is determined based on the fitness values of the previous generation's chromosomes. Individuals with better quality, i.e., shorter path lengths from the starting vertex to the ending vertex, have a higher probability of being selected.

5.2.4 Crossover and Mutation Operations

The process of crossover involves two selected chromosomes. First, a random number is generated to determine the crossover point, which represents the position in the chromosome where gene exchange occurs. Partial gene exchange is performed at this position. Mutation involves reversing a certain gene in the chromosome, changing it from 1 to 0 or vice versa. So, it is possible for better solutions generated in previous generations to be lost during the genetic process, which slows down the attainment of the optimal solution. To improve the convergence of the genetic algorithm and obtain better solutions as quickly as possible, a self-learning process for individual optimization is introduced during the mutation operation. After a gene mutation, the fitness value of the newly generated chromosome is calculated. If the fitness value is smaller, indicating a shorter path, it is retained; otherwise, the original solution remains unchanged. If there is no improvement in the solution for N/3 consecutive times, where N represents the number of vertices from the starting vertex to the ending vertex, the process ends.

5.3 Minimum Spanning Tree Construction Process (MST)

Unweighted graphs can be transformed into weighted graphs from the perspective of real estate developers using the model from the first question. The genetic algorithm is then used to optimize Prim algorithm for finding the MST of the weighted graph.

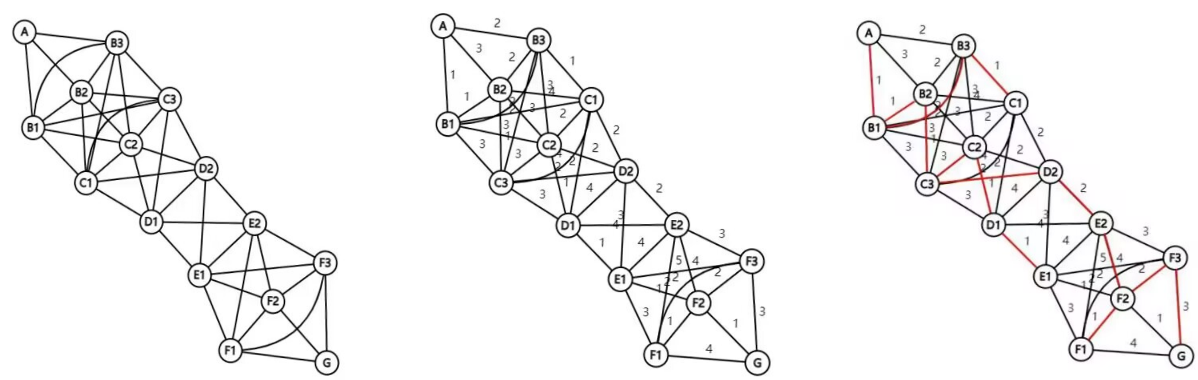


Figure 8: Process of MST

The genetic algorithm for solving the MST problem is as follows: Select $p(h)$ from $p(n-1)$ represents selecting a set of parents from the previous generation population for crossover and mutation operations, where $p(n)$ represents the nth generation population. Generate $[p(n)]$ represents the initial generation of a population at the start of the program. Evaluate $[p(n)]$ represents the computation of the fitness of each individual. In the initial few iterations, individuals with varying qualities and low fitness appear in the genetic algorithm. As the number of iterations increases, individuals with higher fitness will be selected through the process of natural selection.

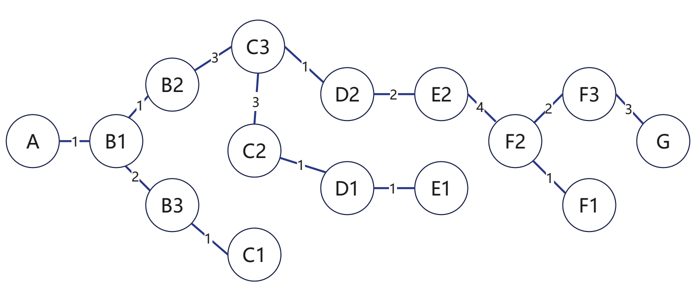


Figure 9: Process of MST

6 Landmark Preservation Model

6.1 Overview

In areas with high disaster risks, Insurance companies have the option to decline underwriting if they consider the risks to be too high. However, community leaders must make informed decisions to determine whether to protect buildings of significant value within the community.

We developed a comprehensive evaluation system for assessing the value of landmarks. Because of the distinctiveness of landmarks, evaluating their value necessitates considering numerous unquantifiable grey indicator data. Therefore, we have opted for a comprehensive evaluation system based on the COWA-Grey Clustering method, which employs the COWA weighting method to determine the weights of evaluation indicators, and then utilizes grey clustering analysis to derive the value index of landmarks. [3].

Furthermore, to aid leaders in determining suitable protection strategies based on the value index of landmarks, we introduce the GE matrix commonly used in business analysis for Value-Cost analysis of landmarks. The flow chart of our process is shown in 10.

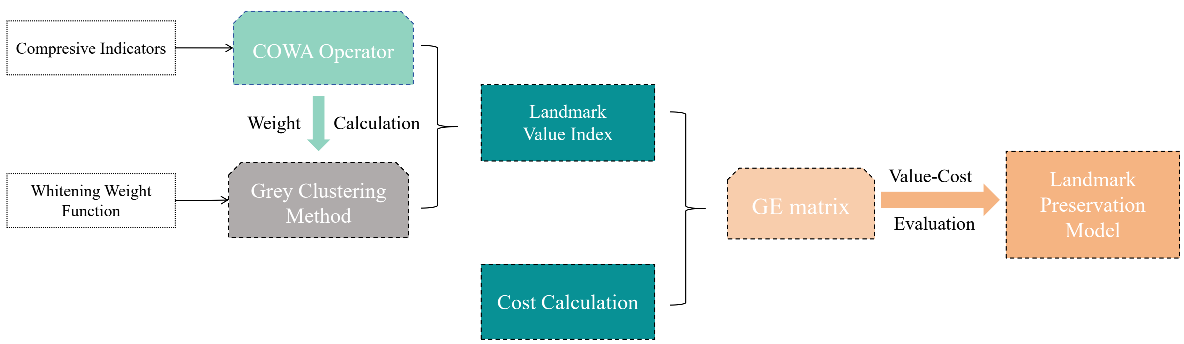


Figure 10: Flow chart of Landmark Preservation Model

6.2 Landmark Value Index Calculation

6.2.1 Selection of Indicators

As an important component of the community, the value of landmark buildings is often determined by multiple factors. The evaluation of landmark buildings often reflects their comprehensive value based on their characteristics and influencing factors, combined with subjective and objective evaluation methods. Based on relevant research on the preservation of historic buildings, we have established a comprehensive evaluation system. We have selected four primary indicators: culture value, history value, economy value, and community value, and corresponding secondary indicators.

- **Culture Value**

- Religion and Belief(RV)
  Buildings with religious significance serve as carriers of human religious and cultural heritage and unique cultural landscapes.[5] The religious nature of landmark buildings is an important part of their cultural value, such as the Notre-Dame Cathedral in Paris.

- Aesthetic and Art(AA)
  The aesthetic value of buildings is generally reflected in their artistic beauty, including design style, ethnic and regional characteristics, architectural decorations, and artistic elements of architectural form.

- Scarcity and Uniqueness(SU) As a landmark building, it is often the most representative symbol of the region and possesses uniqueness. However, we also need to consider whether a landmark building is irreplaceable.

• **Historic value**

- Land Mark Age(LA)
  The older the construction date, the higher its historical value. It is not only a witness of history but also a firsthand experience of history.

- Events and People(EP)
  The association with significant historical events and figures also reflects its historical value. For example, the Confucius Temple in China commemorates the great Chinese educator Confucius and holds great historical significance.

- Antique and Heritage(PH)
  Landmark buildings with historical significance can also be used to reflect the aspects of various historical periods, social systems, and social production and life. These historical relics have archaeological value and historical evidential value.

- Documented in Literature(DL)
  If a landmark building is recorded in literature, it indicates that it has certain value for historical research. For example, the Bukkoji in China reflects the achievements of contemporary Chinese wooden structure architecture.

• **Economic Value**

- Tourism Industry (TI)
  A landmark building can promote the development of tourism centered around itself, thereby stimulating the local economy. For example, the Cape Hatteras Lighthouse attracts over 3 million visitors annually and has high tourism value.

- Business Value(BV) Landmarks can drive the development of surrounding areas, such as business districts, promoting local economic growth. Additionally, some landmark buildings themselves have commercial functions, such as the Twin Towers in New York.

- Employment Opportunities(EO) The economic value of landmark buildings not only directly promotes local industrial development. This helps alleviate employment market pressures and promotes sustainable community development.

• **Community Value**

- Spirit and Emotion(SE)
  As a landmark building, it may form a close emotional connection with local residents during its construction and existence, influencing their general values. The sense of ownership and identification of the landmark by community residents reflects its profound spiritual value.

- Education Value(EV)
  As a cultural benchmark, landmark buildings can collaborate with schools to promote the development of community education.

- Landscape Value(LV)
  The harmony between a landmark and the surrounding community landscape and its impact on community layout also represent its value.

6.2.2 Weight Determination

1. The COWA operator weighting method assigns weights based on the combination numbers, arranging the expert scores in order:

$$\left[a_0, a_1, a_2, \cdots, a_j, \cdots, a_{n-1}\right]$$

2. Calculate the absolute weight of the indicator Let the weight of the data a_j be $\theta_j$and give it weight:

$$\theta_{j+1} = \frac{C_{n-1}^{j}}{\sum_{j=0}^{n-1} C_{n-1}^{j}} (j = 0, 1, 2, \cdots, n-1). \tag{17}$$

3. Combined with the ranking data a_j the absolute weight value of each index factor is obtained

$$\omega_i = \frac{\overline{\omega}_i}{\sum_{i=1}^{m} \omega_i} \left(i = 1, 2, \cdots, m\right) \tag{18}$$

.

   where m is the number of indicator factors.

4. Calculate the relative weights of each indicator

$$\omega_i = \frac{\overline{\omega}_i}{\sum_{i=1}^{m} \omega_i} \left(i = 1, 2, \cdots, m\right) \tag{19}$$

.

According to our calculation results, the weighted indicator system is shown in the figure (cited figure). Among them, the weight of the first-level index is 0.276 for historical value, 0.272 for cultural value, 0.218 for economic value, and 0.233 for community value. And the corresponding weight value of the secondary index was determined. Of course, what we're giving here is a universal weight, and depending on the nature of the place and landmark, we can still use our model to calculate a more personalized weight value.

Goal	First-Level Indicators	Weights	Second-Level Indicators	Weights
The Value of Land Mark	History	0.276	Documents	0.195
			Age	0.202
			Events	0.259
			Heritage	0.343
	Culture	0.272	Religion	0.285
			Aesthetics	0.342
			Scarcity	0.373
	Community	0.233	Education	0.238
			Landscape	0.349
			Spirit	0.413
	Economic	0.218	Employment	0.31
			Business	0.315
			Travel	0.375

Figure 11: The Weights of Each Indicator

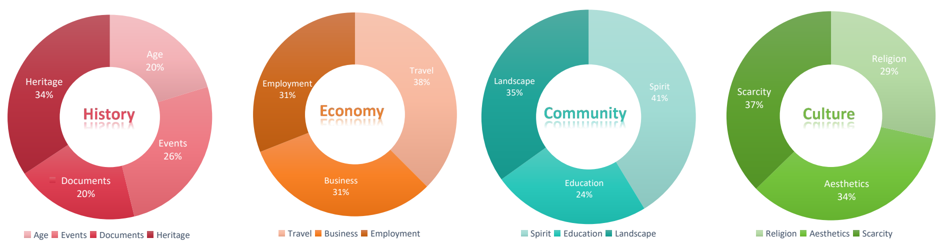


Figure 12: Flow chart of our work

6.2.3 Landmark Value Index Calculation

The value evaluation of landmark buildings is a multi-index comprehensive evaluation system, which needs to evaluate indicators of different properties that are difficult to quantify. We choose the gray clustering method to evaluate each index, and realize the transition of information from white state (unknown) to black state (known) through the whitening weight function.

Determine the evaluation gray category There is no unified standard for the evaluation of the value of landmarks, and the assessment of the value of landmarks should be adapted to local conditions. Combined with the actual problem and the Grey Cluster evaluation theory[1], we divided the landmark value evaluation grades into four grades: extremely high, high, medium and low and assigned values.

Table 4: Grey Cluster Level

Level	extrmely high	high	medium	low
Sore Range	$[10, 8)$	$[8, 6)$	$[6, 3)$	$[3, 0]$

Establish the gray fuzzy evaluation matrix

Invite experts to score and build an initial sample matrix $\boldsymbol{D}_i = \left|d_{ijk}\right|_{s\times p}$.
d_{ijk} represents the score of the $k-th$ expert on the second-level indicator R_{ij}, $k = 1, 2, ..., p$; s is the number of evaluation indicators R_{ij} listed in the matrix D_i.The gray evaluation weight value

of the $e-th$ evaluation gray class is solved, and the gray fuzzy evaluation matrix is obtained.

This process is shown in the pseudocode below.

Establish comprehensive evaluation vector

The clustering method was used to evaluate the secondary index by combining the weight ω_i of the secondary index. The results of this formula are used to calculate the first-level index evaluation.

$$M_i = \omega_i \times Z_i \tag{20}$$

Contain a comprehensive evaluation matrix

$$M_0 = [M_1, M_2, M_3, M_4]^{\mathrm{T}}. \tag{21}$$

Calculate the comprehensive evaluation vector of the first-level index

$$Y = \omega_0 \times M_0 \tag{22}$$

ω_0 represents the weight of first-level indicator. **Caculate the Landmark Value Index**

To reduce the information loss caused by the traditional principle of maximum weight to determine the gray class, the comprehensive evaluation vector and the threshold U are synthesized to carry out single-value processing.

$$W = Y \times U^T \tag{23}$$

Algorithm 1 Grey Fuzzy Matrix Calculation (Part)

Input: D - A matrix of evaluation data
Output: X - Total grey relational grade for each option, Z - Grey fuzzy matrix
Initialize Z, a matrix of size $m \times n$, where m is the number of options
Initialize X, an array of size m
for each option i from 1 to m **do**
    Calculate $Z[i, 1]$ using the first whitening weight function for all elements in $D[i, *]$
    Calculate $Z[i, 2]$ using the second whitening weight function for all elements in $D[i, *]$
    Calculate $Z[i, 3]$ using the third whitening weight function for all elements in $D[i, *]$
    Calculate $Z[i, 4]$ using the fourth whitening weight function for all elements in $D[i, *]$
    Sum the values in $Z[i, *]$ to get the total grey relational grade $X[i]$
    if $X[i] \neq 0$ **then**
        Normalize the row $Z[i, *]$ by dividing each element by $X[i]$
    end if
end for
return Z

6.3 Landmark Preservation Scheme

We calculate the comprehensive cost of maintaining the landmark, and visualize it through the GE matrix based on the comprehensive cost combined with the value.

6.3.1 Cost calculation

For different landmarks and different maintenance schemes, the corresponding costs are different. In order to conduct a comprehensive evaluation plan before proceeding with the option selection, we need to calculate the maintenance cost of the various protection options to help us make better decisions. The cost is calculated as follows:

$$COST = \eta_i Age + \eta_i Labor + \eta_i Transportation + \eta_i Material + \eta_i DRI + \mu \tag{24}$$

Age represents the age of the landmark, Labor represents the labor cost of the landmark, Transportation represents the transportation cost of the landmark, Material represents the cost of the materials required by the landmark, and DRI represents the risk index of the landmark.

$$Labor = P_{labor} \times n_{labor} \tag{25}$$

$P_L abor$ represents the average price of labor in the local area, and n represents the amount of labor required.

$$Transportation = P_{tran} \times d_{tran} \tag{26}$$

$P_t ran$ stands for the average price of local transportation costs; $d_t ran$ represents the estimated transportation distance.

$$Material = P_{mat} \times V_{mat} \tag{27}$$

$P_m at$ represents the unit price of the material required for maintenance, and $V_m at$ represents the mass or volume of the material required. μ is the cost fluctuation factor. Based on this formula, we can calculate a rough estimate of the cost to better assist us in formulating the plan.

6.3.2 Value-Cost Evaluation

We introduce the GE matrix to visualize the Value-Cost comprehensive evaluation of the landmark.

The GE matrix, also known as the McKinsey Matrix, is a strategic management tool used to evaluate a business portfolio. The GE matrix can be applied in several fields as an analysis and decision tool.

According to the different cases of Value-Cost Evaluation, we divide the GE matrix into nine parts. to identify different protection strategies. The criteria for division are shown in Table 5

The $Value_m in$, $Cost_m in$, $Value_m ax$, and $Cost_m ax$ here represent the world average level,

Table 5: Level Classification of Value and Cost

Value	Level	Cost	Level
$Value_m in$ $0.3Value_m ax$	Low	$Cost_m in$ $0.3Cost_m ax$	Low
$0.3Value_m ax$ $0.6Value_m ax$	Medium	$0.3Cost_m ax$ $0.6Cost_m ax$	Medium
$0.6Value_m ax$ $Value_m ax$	High	$0.6cost_m ax$ $0.6Cost_m ax$	High

which is the average value calculated by us based on the existing landmark protection strategy data and data, which is representative and universal.

Landmarks fall at different locations in the GE matrix, representing different protection schemes. To simplify the analysis, we divided the GE matrix into three levels based on numerical values.

Theoretically, when the landmark is located in the red zone, its cost and value are at a high standard, which means that we need to develop more comprehensive and integrated protection measures. It is necessary to consider the economic interests while also attaching great importance to the value of the building itself. When necessary, we can implement a relocation strategy to preserve the overall value of the building.

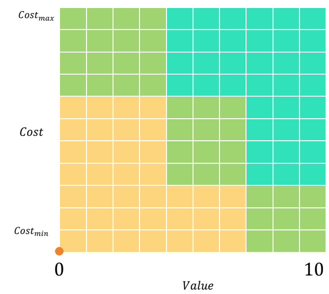


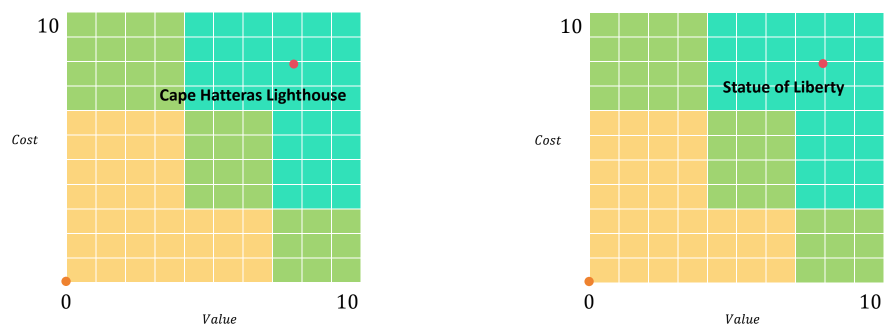
Figure 13: GE matrix

When in a gray area, landmarks have a moderate V-C ratio. We can choose the method of in-situ restoration, combined with the construction of defense facilities to resist disasters.

When located in the Blue Zone, we can opt for a more conservative maintenance strategy. This conservative approach can be done on a small scale without having to overinvest in the landmark's resilience.

6.3.3 Case Study

Based on the U.S. research and protection of endangered landmarks, we can apply the GE Matrix in the Cape Hatteras Lighthouse and the Statue of Liberty. The results are shown in 14



(a) Cape Hatteras Lighthouse (b) Statue of Liberty

Figure 14: data

By applying our landmark protection model, it was calculated that the landmark value index of the lighthouse is 7.695 and that of the Statue of Liberty is 7.967, which has a higher value.

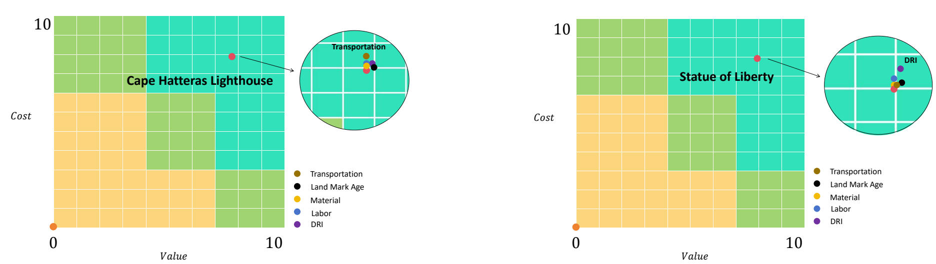
- **Lighthouses**: In our GE matrix, lighthouses should be protected with high inputs. Since its construction in 1870, the lighthouse has become one of the most famous landmarks in the area. The unique appearance of the black and white stripes attracts millions of tourists. It was able to warn ships up to 22 miles away of the dangers of nearby Diamond Shoal. However, in the face of rising sea levels and increasingly severe hurricane hazards, lighthouses are at risk of being submerged. Therefore, we need to develop high-input protection measures to protect their value. Lighthouse has already taken relocation measures, which is consistent with our analysis.

- **Statue of Liberty**: Conservation measures for the Statue of Liberty should be high-investment conservation, with the option of relocation if necessary. The Statue of Liberty is a representative landmark of the United States, symbolizing the common values pursued by the American people. At the same time, it is also listed as a world cultural heritage and has a high-value index. However, in recent years, hurricane attacks, floods, and rising sea levels have all posed a huge threat to it. Therefore, we need to maintain it regularly and invest enough money.

Community leaders should build on this model and conduct public opinion polls to gain broad support from the community. Raise people's awareness of protection and risk. At the same time, it chose to reach reinsurance cooperation with insurance institutions to reduce the pressure of landmark protection in the face of catastrophes.

7 Sensitive Analysis

In our landmark conservation model, the determinant of cost is . Based on the cost formula, we perform sensitivity analysis on the important variables by changing the indicators. When one indicator changes, the other four should remain the same. The five indicators were changed alone from 0.001 to 0.05. We implemented this process of change for two landmarks, the Statue of Liberty and the Lighthouse.



(a) Cape Hatteras Lighthouse (b) Statue of Liberty

Figure 15: Sensitive Analysis

The 15 shows how the process has changed.

- **For the Statue of Liberty**, its cost is sensitive to disaster risk, labor, and age of the building. This is because of the technical aspects. The Statue of Liberty was built in 1886, but the aging of the building increased the cost of repairs. Besides, in the aftermath of Hurricane Sandy, the Statue of Liberty is in great danger. Increasing climatic hazards put them at risk of requiring long-term monitoring and repeated repairs. For New York, transportation is convenient, and its maintenance costs are not sensitive to transportation.

- **For lighthouses**, its cost is most sensitive to traffic factors. Lighthouses have been displaced by rising sea levels and are not sensitive to current disaster risks, but the risks they face remain. In addition, the lighthouse is mainly built of bricks with fixed materials. The labor cost is relatively stable. Due to the increasing disaster risk, Highway 12, the main route along the Bund in the area, is facing the risk to be destoryed.

8 Model Evaluation and Further Discussion

8.1 Strengths

- Our model collects extensive and comprehensive data, which facilitates our comprehensive analysis. We also used ArcGIS for data visualization to make our results more intuitive.

- In solving the insurance program problem, we chose a logistic regression model to make a reasonable prediction of whether the insurance company will assume the policy, which is highly accurate and realistic.

- In solving the problem of site selection for real estate companies, we use the PRIM algorithm improved by genetic algorithm to determine the best site selection plan. Genetic algorithm is an optimization algorithm based on the principle of biological evolution with high parallelism and global search capability.

- When evaluating the landmark value, we chose the combined solution method of COWA-Grey Clustering Method, which reduces the limitation of expert scoring to a certain extent.

8.2 Weakness

- For the real estate site selection model, we consider the demographic factor, but the consideration of the dynamically growing population is not comprehensive.we can add a logistic blocked growth model to the original population factors.

- Our landmark value index still needs to be scored by experts, and the quantitative indicators are not sufficiently considered.we can introduce some quantitative indicators to assist experts in comprehensive evaluation.

9 Conclusion

Catastrophe risk is rising, and human society is facing a crisis. Reaching a deep cooperation with insurance companies can help us better survive these catastrophes. How to reach a win-win situation and maintain sustainable development is our common goal.

Through our model, we can rationalize the insurance company's catastrophe policy. It can also assist real estate developers in making rational site selections. To develop a reasonable protection strategy in communities with very high risk, we can calculate the landmark value index to measure its importance level and select the most reasonable protection plan through cost analysis.

All in all, it is our common responsibility to protect the public property of mankind. Everyone is responsible for facing catastrophes. We need to work with insurance companies and governments to achieve a "handshake transition" and maximize the risk of future catastrophes.

Please pay attention to the protection of St. Mark's Basilica

Dear Community Leaders and Residents,

I am writing to discuss our precious historical landmark, St. Mark's Basilica in Venice. Flooding and rising sea levels continue to cause great disaster and put our city and landmark in danger. For your, our community leaders, had to consider how to protect this precious historic landmark.

We used a comprehensive IPFM and LPM and concluded that St. Mark's Basilica's comprehensive in-situ preservation is of critical importance to our community.To protect this valuable landmark , we have considered the "MOSE" program carried out locally in Venice and have made the following recommendations for St. Mark's Basilica.

Self-maintenance aspects:

- Establishment of protective measures: Investing in appropriate protective measures embankments, flood walls, damp-proof coatings.
- Regular maintenance: Inspection and repair of possible structural problems and damage.
- Elevation of Venice: Reference to Prof. Gamborati of the University of Padua proposed the elevation of Venice by about 25-30 centimeters by injecting water into the ground.

Cost aspects:

- Consider reinsurance: We recommend working with insurance companies to explore reinsurance strategies to ensure that St. Mark's Cathedral is properly compensated in the event of a disaster.
- Community Involvement: Encourage community participation in the protection and maintenance of St. Mark's Cathedral. Schools and communities should increase their awareness. The benefits of preserving our landmarks outweigh the drawbacks in the long run.

2025 MOSE Program Implementation

2026 Regular maintenance Insurance program preview

2030 Completion of protective measures

2035 Reinsurance agreements reached

Last but not least, we hope to work with you and the entire community to maintain the sustainability and long-term preservation of St. Mark's Cathedral. If you have any questions or suggestions, please feel free to contact us.

Thank you for your interest in and support of our community's landmark heritage.

Sincerely yours,

A citizen who wished to remain anonymous

References

[1] Xiao Liu Hui Yu. Apprasising method of comprehensive value of historic buildings based on grey clustering method. *Sichuan Building Science Research*, 35(240-242), 2009.

[2] Mingbo Liu. The study on mechanism design of catastrophe risk financing in china——base on public-private partnership perspective. 2014.

[3] Fajia Shan Xuexin Zhu, XianMei Chen. Evaluation of sustainable renewal of historic district based on ism-grey clustering method. *Journal of Hunan City University(Natural Science)*, 32(49-56), 2023.

[4] Yule Yan. Commercial complex site selection based on gis and random forest research – taking ganzhou city as an example. 2023.

[5] Fan Zhang. Strengthening the protection of religious architectural heritage and promoting cultural mixture and inheritance. *Chinese Religion*, (48-49), 2023.

Report On Use of AI

1. OpenAI *ChatGPT(Nov 5,2023 version,ChatGPT-4,)*
Query1:*How to view the United States country map containing various information*
Output:*To view a detailed and interactive map of the United States containing various information, you have several options: 1. **Online Mapping Platforms:***
*- **Google Maps:** Google Maps provides a comprehensive view of the United States with detailed street maps, satellite imagery, and the ability to explore different locations. You can also search for specific places and get information about businesses and points of interest.*
*- **OpenStreetMap:** OpenStreetMap is a collaborative mapping platform that allows users to view and edit maps. It provides a wide range of layers, including streets, topography, and satellite imagery. Various applications and websites use OpenStreetMap data for mapping purposes.*
*2. **Geographic Information Systems (GIS) Websites:***
*- **ArcGIS Online:** Esri's ArcGIS Online platform offers a wealth of geospatial data and maps. You can explore pre-made maps or create your own using a variety of layers, including demographic data, land use, and environmental information.*
*- **USGS Earth Explorer:** The U.S. Geological Survey (USGS) Earth Explorer allows you to access and download satellite and aerial imagery, as well as various geospatial datasets.*
*3. **Government Websites:***
*- **U.S. Census Bureau:** The U.S. Census Bureau provides interactive maps with demographic data, including population density, income, and more. The "Census Explorer" tool is a useful resource.*
*- **National Oceanic and Atmospheric Administration (NOAA):** NOAA offers various maps related to weather, climate, and environmental conditions.*
*4. **Data Visualization Platforms:***
*- **Tableau Public:** Tableau Public is a data visualization platform that allows users to create interactive maps. You can find and explore public datasets or upload your own data for mapping.*
*5. **Educational and Research Institutions:***
*- **National Geographic MapMaker Interactive:** National Geographic provides an interactive map tool that allows users to explore different layers of information, including physical geography, human geography, and more.*
*6. **Weather and Environmental Websites:***
*- **The Weather Channel:** Weather websites often include interactive maps with real-time weather conditions, radar imagery, and forecasts.*
*7. **Mobile Apps:***
*- **Map Apps:** Many map apps on smartphones and tablets provide detailed views of the United States. Examples include Google Maps, Apple Maps, and other third-party map applications.*
*8. **Educational and Research Institutions:***
*- **University Research Centers:** Some universities and research institutions offer online mapping tools and resources related to specific topics, such as land use, environmental studies, or urban planning.*
When exploring these platforms, consider the specific type of information you are interested in, such as demographic data, environmental conditions, land use, or weather. Each platform may provide different layers of information, allowing you to customize your view of the United States based on your interests and needs.

Query2:*Name some landmarks that have a similar value to the Cape Hatteras Lighthouse and have been affected by experiences extreme weather events.*

Output: *1. **Statue of Liberty, United States:***

- Extreme Weather: Hurricane Sandy (2012)

- The Statue of Liberty, located in New York Harbor, was temporarily closed and suffered damage during Hurricane Sandy. The storm surge and flooding affected infrastructure on Liberty Island.

*2. **Venice, Italy:***

- Extreme Weather: Flooding

- The historic city of Venice experiences periodic flooding, known as "aqua alta," due to a combination of high tides, storm surges, and rain. The flooding poses a threat to the city's architecture and cultural heritage.

*3. **Tower Bridge, United Kingdom:***

- Extreme Weather: Winter Storms

- London's iconic Tower Bridge has faced closures and disruptions during severe winter storms, impacting transportation and public access.

*4. **Machu Picchu, Peru:***

- Extreme Weather: Heavy Rainfall and Landslides

- Machu Picchu, a UNESCO World Heritage Site, has faced challenges related to heavy rainfall and landslides in the surrounding Andean region, affecting access and conservation efforts.

*5. **Petra, Jordan:***

- Extreme Weather: Flash Floods

- The ancient city of Petra has experienced flash floods, posing a risk to its archaeological structures. In recent years, efforts have been made to address the impact of heavy rainfall on the site.

*6. **Mont Saint-Michel, France:***

- Extreme Weather: Tidal Surges and Storms

- The island commune of Mont Saint-Michel faces the risk of tidal surges and storms. Measures have been taken to protect the site from erosion and flooding.

*7. **Taj Mahal, India:***

- Extreme Weather: Air Pollution and Acid Rain

- The Taj Mahal has been affected by air pollution and acid rain, impacting the integrity of its marble facade. Conservation efforts are ongoing to mitigate the effects of environmental factors.

*8. **Acropolis of Athens, Greece:***

- Extreme Weather: Heatwaves and Pollution

- The Acropolis has faced challenges due to heatwaves and pollution in Athens. Conservation efforts aim to protect the ancient structures from environmental stressors.